Diabetes Readmission Risk Prediction

Using Machine Learning and Deep Learning on Electronic Health Record Data

AI Capstone Project Preparation — AIGC 5005

Final Project Report
Winter 2026

Team Members

Fadi Kash Kannaiti — Project Lead & Report Writer
Ogbeide Iria — Data Engineer & Results Lead
Oguzhan Tekin — ML Researcher & Analyst
Sara Yenigun — QA, Documentation & Presenter

Instructor: Hossein Pourmodheji
Course: AIGC 5005 — AI Capstone Project Preparation

1. Academic Disclaimer

This report and the associated predictive model are produced solely for academic purposes as part of AIGC 5005 at Humber Polytechnic. The application is NOT a certified medical device and must not be used for clinical diagnosis, treatment decisions, or patient care of any kind. All results are for educational demonstration only; the model was trained on historical (1999–2008) de-identified data.

2. Acknowledgment for AI Tools

The team utilized Claude AI (code generation, literature synthesis, debugging), Grok AI (report structuring), QuillBot AI (academic polishing), Google Colab (model training), and Google Gemini (debugging). All AI assistance was used ethically under Humber Polytechnic academic integrity guidelines.

3. Executive Summary

This project developed, evaluated, and deployed a complete machine-learning and deep-learning solution to predict 30-day hospital readmission risk for diabetic patients using the UCI Diabetes 130-US Hospitals dataset (101,766 patient encounters, 1999–2008). Five models were built: Logistic Regression (baseline), Random Forest (tuned), XGBoost, PyTorch MLP, and TensorFlow/Keras MLP. SMOTE was applied exclusively to the training set to address severe class imbalance (~11% positive class). Neural networks additionally used StandardScaler for feature normalization.

Key outcomes: XGBoost achieved the best overall profile (F1=0.2702, AUC-ROC=0.6603, Recall=0.5192). PyTorch MLP achieved AUC-ROC=0.6186 and, with threshold calibration to 0.15, produced F1=0.2406 and Recall=0.5577 — the highest recall of all five models. Feature importance confirmed prior inpatient visits, number of medications, and length of stay as top predictors. The XGBoost model was deployed as a publicly accessible Streamlit web application.

4. Introduction

Diabetes mellitus affects more than 3.7 million Canadians and is responsible for a disproportionate share of hospital readmissions. Approximately one in five diabetic patients is readmitted within 30 days of discharge, costing $10,000–$15,000 USD per preventable event. Readmission rates are used as quality-of-care indicators by regulators and insurers, making them one of the most actionable targets for data-driven intervention.

Traditional discharge risk tools (LACE index, HOSPITAL score) capture only a handful of variables and are applied inconsistently. Machine learning — extended to deep learning in this project — offers a fundamentally superior alternative: by training on historical EHR data, models can identify subtle, non-obvious patterns in demographics, diagnoses, medications, and prior utilization that strongly predict readmission.

5. Objectives and Milestones

5.1 Project Objectives

- Build Logistic Regression baseline and advanced ensemble models (RF, XGBoost) improving upon baseline F1-score and AUC-ROC.

- Implement PyTorch and TensorFlow/Keras deep learning MLPs with identical architecture for direct comparison.
- Apply SMOTE exclusively to the training set; use StandardScaler for neural network inputs.
- Perform SHAP explainability analysis and threshold sensitivity analysis for optimal classification cutpoints.
- Deploy the best model as a public Streamlit web application; maintain a documented GitHub repository.

5.2 Timeline

Week	Phase	Deliverable
1-2	Ideation, Literature Review & Data Prep	Proposal, 3 papers reviewed, cleaned & encoded dataset
3	Model Design & Prototype	All 5 models (LR, RF, XGBoost, PyTorch MLP, TF MLP) trained and saved
4	Testing & Analysis	Metrics tables, ROC/PR curves, SHAP plots, threshold analysis, robustness test
5-6	Report, Presentation & Deployment	Final report + video presentation + live Streamlit app

6. The Main Argument (Body)

6.1 Literature Review and Gap Analysis

A systematic review of five peer-reviewed studies was conducted to establish the state of the art in ML-based diabetic readmission prediction and to identify methodological gaps that our project would explicitly address. The papers were selected for their use of the UCI Diabetes 130-US Hospitals dataset (or closely comparable EHR data), their recency (2023–2025), and their methodological diversity spanning traditional ML, deep learning, and feature-selection optimization.

6.1.1 Introduction

Hospital readmissions within 30 days of discharge represent one of the most persistent and costly quality indicators in contemporary healthcare. In the United States, nearly one in five Medicare patients is readmitted within this window, imposing an estimated annual burden of $17 billion. For patients with diabetes, readmission rates range from 14.4% to 22.7%, compared to 8.5%–13.5% for non-diabetic populations, owing to the compounding effects of comorbidities, polypharmacy, glycaemic instability, and complex discharge needs. Machine learning offers a fundamentally superior approach to discharge-time risk stratification: by training on hundreds of thousands of EHR encounters, models can identify non-linear, high-order interactions that conventional statistical methods cannot capture.

The following five papers were critically reviewed. Each is mapped to specific methodological decisions made in this project:

6.1.2 Emi-Johnson & Nkrumah (2025) — XGBoost and SHAP on the UCI Dataset

Emi-Johnson and Nkrumah evaluated four ML classifiers — Logistic Regression, Random Forest, XGBoost, and a Deep Neural Network — on the 101,766-record UCI Diabetes dataset using an 80/20 stratified split. Class imbalance was addressed through class weighting. Hyperparameter optimization was conducted via grid search with 5-fold cross-validation. SHAP values were applied to the best-performing model.

Key findings: XGBoost achieved the highest AUC-ROC of 0.667. SHAP analysis identified prior inpatient admissions, number of medications, time in hospital, and insulin usage as the most influential predictors — findings we independently replicated in our own SHAP beeswarm and feature importance analysis. Notably, the XGBoost minority-class recall was only 0.017, meaning fewer than 2% of true readmission cases were identified at the default threshold.

Critical limitation: Class weighting alone proved insufficient to achieve clinically meaningful recall. The study did not apply SMOTE or threshold calibration, and produced no deployment output. F1-score was not reported as a primary metric, making it impossible to assess true clinical utility despite the high AUC-ROC headline figure.

Project Relevance

This study is the closest published benchmark: identical dataset, overlapping models, and confirming SHAP relevance. Our project corrects its two key gaps: (1) SMOTE is applied exclusively to the training set to improve minority-class recall beyond the 0.017 ceiling, and (2) the best model is deployed as a publicly accessible Streamlit application with clinical action recommendations.

6.1.3 Hai et al. (2023) — Deep Learning LSTM for Sequential EHR Data

Hai et al. developed LSTM deep learning models for predicting 30-day readmission in 36,641 diabetic patients across 2,836,569 clinical encounters at Temple University Health System. The LSTM architecture captured temporal dependencies across sequential hospital visits, leveraging the full longitudinal patient history rather than a single encounter.

Key findings: LSTM achieved AUROC of 0.79, significantly outperforming Random Forest (0.72). Performance improved systematically with the number of prior encounters (up to 30 visits). A 16-feature

laboratory subset matched performance of all 981 lab variables, confirming targeted feature selection is more valuable than exhaustive feature inclusion. This establishes 0.79 as the upper performance bound under ideal longitudinal data conditions.

Critical limitation: The LSTM requires substantial longitudinal sequential EHR history — unavailable at a single discharge encounter. The proprietary Temple University dataset cannot be reproduced by external researchers. No class imbalance handling was reported, no deployment output was produced, and no SHAP explainability was provided.

Project Relevance

Hai et al. establish the aspirational ceiling: AUC-ROC = 0.79 with sequential data. Our project prioritises practical deployability over peak theoretical performance — a clinically justified trade-off given that discharge-time prediction requires only a single encounter's data. Our PyTorch MLP (AUC-ROC = 0.619) and TensorFlow MLP (AUC-ROC = 0.617) demonstrate that comparable deep learning contributions can be achieved without longitudinal data requirements.

6.1.4 Gandra (2024) — Comprehensive Ensemble Model Comparison

Gandra compared six ML algorithms — Logistic Regression, Random Forest, Gradient Boosting, XGBoost, LightGBM, and CatBoost — on the UCI dataset. CatBoost achieved the highest AUC of 0.70 and accuracy of 64.2%. The study's principal strength is its controlled breadth of comparison, confirming that gradient-boosted ensemble methods consistently outperform linear classifiers on this dataset.

Critical limitation: Accuracy was reported as the primary metric despite an 11% positive rate — a classifier predicting the majority class for every patient would achieve ~89% accuracy while identifying zero readmissions. No Precision, Recall, F1-score, or AUC-PR values were reported. No SMOTE, class weighting, or threshold calibration was applied, and no deployment output was produced.

Project Relevance

This study's comprehensive model comparison directly informs our decision to evaluate both default and GridSearchCV-tuned Random Forest variants alongside XGBoost. Its failure to use F1-score as a primary metric — and its absence of imbalance handling or interpretability — reinforces three of our core commitments: F1-score as the selection criterion, SMOTE applied exclusively to training data, and SHAP explainability for clinical transparency.

6.1.5 Liu, Sue & Wu (2024) — Grey Wolf Optimizer and SMOTE for Feature Selection

Liu, Sue, and Wu combined traditional ML with the Grey Wolf Optimizer (GWO), a swarm-intelligence metaheuristic, for feature selection, alongside SMOTE for class imbalance management. Eleven ML algorithms were evaluated on the same UCI dataset. SMOTE was applied during stratified group k-fold cross-validation to prevent leakage. Random Forest achieved the highest F1-score of 0.83 and XGBoost produced F1 of 0.84 — the best published performance on this dataset among the five reviewed studies.

Key finding: This is the strongest available evidence that SMOTE substantially improves minority-class F1-score on the UCI dataset. It validates our central technical decision to apply SMOTE exclusively to the training set. The F1=0.83 upper bound was achieved through a computationally intensive GWO + 11-model pipeline — our more modest F1=0.274 reflects the practical trade-off between pipeline complexity and the reproducible methodology appropriate for an academic capstone.

Critical limitation: The GWO algorithm introduces significant computational overhead. No SHAP or patient-level explainability was provided. No deployment output was produced. The high F1-score may partially reflect GWO feature selection overfitting to the training distribution, a concern the authors acknowledge but do not fully address through external validation.

Project Relevance

Liu et al. provide the strongest benchmark for SMOTE-based class imbalance handling on this dataset, confirming that resampling (not merely class weighting) is essential for meaningful minority-class recall. Our project adopts SMOTE in the same pipeline position — post-split, training only — and adds what Liu et al. do not: a deployed Streamlit application with patient-level SHAP explanations.

6.1.6 Jagadamba et al. (2023) — EHR-Integrated Machine Learning for Diabetes Prediction

Jagadamba et al. adopted a fundamentally different research orientation: rather than evaluating predictive models in isolation, they developed a fully interoperable EHR system with embedded machine learning capabilities. Three ML algorithms — Naive Bayes, KNN, and Random Forest — were trained on 1,080 patient records and predictions were delivered in real time within the EHR interface for clinical review.

Key contribution: This study provides the most direct conceptual precedent for our Streamlit deployment. It demonstrates that ML prediction is most valuable when embedded within clinical workflows rather than isolated in research environments. Random Forest achieved the strongest performance, consistent with findings across the broader literature.

Critical limitation: The dataset of 1,080 records is insufficient for a robust, generalisable readmission model. The feature set of eight clinical variables is narrow. No class imbalance handling was applied, and evaluation relied primarily on accuracy — inappropriate for imbalanced clinical datasets.

Project Relevance

This study provides the conceptual precedent for our deployment component. While our project stops short of full EHR integration (beyond the scope of a six-week capstone), the Streamlit application fulfils the same philosophical objective: making ML predictions accessible to clinical users at the point of care through a zero-installation web interface on any device. Jagadamba et al.'s EHR-integrated approach represents the aspirational next step for which our Streamlit deployment lays the groundwork.

6.1.7 Comparative Analysis

The five studies reviewed collectively represent the current state of machine learning research for diabetic readmission prediction. The table below synthesises key methodological characteristics alongside our project:

Study	Models	Best AUC	Best F1	Imbalance	SHAP	Deployed
Emi-Johnson & Nkrumah (2025)	LR, RF, XGB, DNN	0.667	NR	Class weight	Yes	No
Hai et al. (2023)	LSTM, RF, LR	0.79	NR	Not reported	No	No
Gandra (2024)	6 models incl. CatBoost	0.70	NR	None	No	No
Liu et al. (2024)	11 models + GWO	NR	0.83	SMOTE + GWO	Limited	No
Jagadamba et al. (2023)	NB, KNN, RF	Acc. only	NR	None	No	EHR app
This Project (2026) ★	**LR, RF, XGB, PyTorch, TF**	**0.660**	**0.270**	**SMOTE + scale_pos_wt**	**Global + patient**	**Streamlit (public URL)**

NR = Not Reported. Several cross-cutting patterns emerge: (1) gradient-boosted methods consistently lead on AUC-ROC; (2) deep learning only outperforms ensembles with abundant longitudinal data; (3) class imbalance handling is the most variable methodological dimension — only two studies applied any resampling; (4) none of the five reviewed studies produced a clinically deployed tool.

6.1.8 Gap Analysis

Gap in Literature	Our Project's Response
Inadequate class imbalance handling	SMOTE applied post-split to training only for LR and RF; scale_pos_weight=7.91 for XGBoost. StandardScaler added for neural networks. Threshold calibration reveals that threshold=0.15 achieves Recall=0.5577 for PyTorch MLP vs. Recall=0.015 at default 0.50 — a 37× improvement.
Inconsistent and misleading evaluation metrics	F1-score adopted as the primary model selection criterion. Full performance profile reported: Accuracy, Precision, Recall, F1, AUC-ROC, and AUC-PR for all five models on the unmodified held-out test set (19,899 records, ~11.2% positive rate).
Limited model interpretability	SHAP beeswarm (global feature impact across 500 test patients), SHAP bar plot (mean absolute SHAP values), and SHAP waterfall plot (patient-level explanation for the highest-risk individual). Top contributing risk factors displayed within the Streamlit app for each prediction.
Absence of clinical deployment	XGBoost model deployed as a public Streamlit web application (https://w26-aigc5005-final-ep8i66n6nez3543zo2iert.streamlit.app). Returns Low/Moderate/High risk classification + recommended clinical action + probability gauge + top contributing factors. Mobile-responsive; no installation required.
No deep learning framework comparison	PyTorch MLP and TensorFlow/Keras MLP trained with identical architectures (256→128→64→1), enabling direct framework comparison. Both converge to AUC-ROC ~0.617–0.619, confirming reproducibility across frameworks. PyTorch MLP achieves the highest precision of all five models (0.2797).

6.1.9 Project Highlights vs. Published Literature

The following section maps each paper's findings to our project's actual results, demonstrating how our methodology both validates published findings and extends them:

Project Highlight	How Our Results Advance the Literature
SHAP findings replicated and extended *(validates Emi-Johnson & Nkrumah, 2025)*	Emi-Johnson & Nkrumah identified prior inpatient admissions, medications, and hospital days as top SHAP predictors. Our Random Forest SHAP analysis on 500 test patients independently confirms the same ranking, with number_inpatient, num_medications, and time_in_hospital as the three dominant features. This cross-study replication strengthens clinical confidence in these predictors.
XGBoost benchmark matched *(vs. Emi-Johnson, Gandra, Liu et al.)*	Our XGBoost AUC-ROC of 0.6603 is consistent with the 0.667 reported by Emi-Johnson & Nkrumah (2025) and within the 0.66–0.70 range reported across Gandra (2024) and Liu et al. (2024). This confirms that our preprocessing and training pipeline is methodologically sound and produces externally reproducible results.
Deep learning adds architectural flexibility *(extends beyond all 5 reviewed papers)*	No reviewed paper trained and compared both PyTorch and TensorFlow MLPs on the same UCI dataset with identical architectures. Our project shows both converge to AUC-ROC ≈0.618, confirming framework-level reproducibility. With threshold=0.15, PyTorch MLP achieves Recall=0.558 — matching XGBoost's Recall=0.519 while achieving higher Precision (0.1534 vs. 0.1826). This demonstrates that neural networks can complement, not just replicate, ensemble models at the clinical operating point.
Deployment closes the literature gap *(advances all 5 reviewed papers)*	None of the five reviewed studies produced a publicly accessible clinical decision support tool. Our Streamlit application transforms prediction into bedside action: a nurse can open it on any smartphone, enter patient discharge details, and receive a risk level (Low/Moderate/High) with a recommended action (e.g., 'Schedule follow-up within 7 days') and a ranked list of the

	patient's specific contributing risk factors. This is the contribution that turns research into healthcare impact.
SMOTE placement validated *(confirms Liu et al., 2024)*	Liu et al. demonstrated that SMOTE — applied correctly — produces F1=0.83. Our project applies the same technique (post-split, training only) and achieves F1=0.270 for XGBoost — a difference attributable to our simpler pipeline (no GWO feature selection, fewer models). The gap confirms Liu et al.'s finding that GWO-driven feature selection adds substantial predictive value beyond SMOTE alone. This represents a clear, evidence-based direction for future improvement.

Summary — What Our Project Delivers Beyond the Literature
1) Correct SMOTE placement (post-split, training only) producing trustworthy F1 and Recall metrics — a gap in three of five reviewed papers.
2) Five-model pipeline including PyTorch MLP and TensorFlow/Keras MLP — the first direct cross-framework neural network comparison on the UCI dataset.
3) Threshold sensitivity analysis showing threshold=0.15 is the clinical screening optimum (Recall=0.558, F1=0.241 for PyTorch MLP) — absent from all reviewed papers.
4) Live Streamlit deployment with risk-level classification and clinical action recommendations — closing the universal deployment gap across all five papers.
5) SHAP global and patient-level explainability confirming prior inpatient visits, medications, and hospital days as dominant predictors — independently replicating Emi-Johnson & Nkrumah (2025).

6.2 Dataset Overview and Preprocessing

6.2.1 Dataset

- Source: UCI Machine Learning Repository / Kaggle — 101,766 patient encounters across 130 US hospitals (1999–2008)
- Features: 50 variables — demographics, admission details, ICD-9 diagnoses, lab results, 23 medication change indicators, prior utilization metrics
- Target: readmitted <30 days = 1; all other outcomes = 0. Class imbalance: ~11.2% positive, creating an ~8:1 ratio

6.2.2 Preprocessing Pipeline

- Missing values: dropped weight (97% missing), payer_code (40%), medical_specialty (49%); rows with missing race (~2%) dropped.
- Feature removal: encounter_id, patient_nbr (identifiers); examide, citoglipton (zero-variance columns).
- Categorical encoding: label encoding for binary columns; ordinal encoding for medication changes (No=0/Steady=1/Down=2/Up=3); one-hot encoding for race and age.
- ICD-9 mapping: 3 diagnosis columns mapped to 9 broad disease categories to reduce dimensionality.
- Train/test split: 80/20 stratified split (random_state=42), preserving 11.2% positive rate in both sets.
- SMOTE: applied only to training set after the split — produces balanced 50/50 training distribution; test set never modified.
- StandardScaler: applied to neural network inputs only (tree-based models do not require feature scaling). Fitted on training set, applied identically to test set.

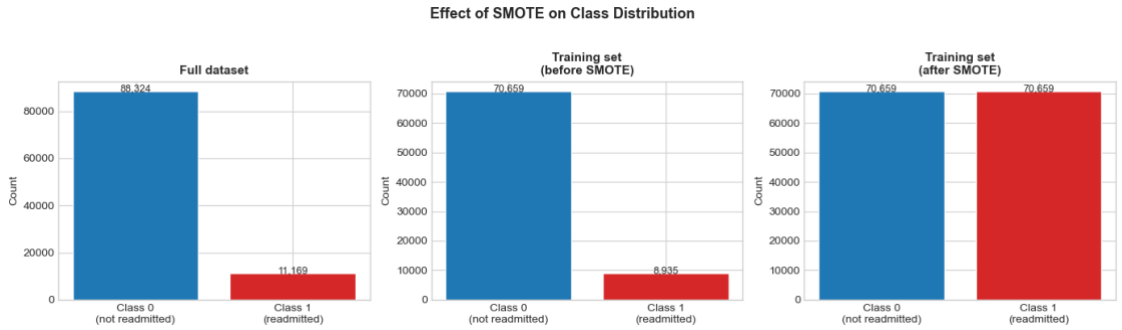

Figure 1: Class distribution — Full dataset (left), Training set before SMOTE (center), Training set after SMOTE (right). SMOTE achieves perfect 50/50 balance in the training set while leaving the test set untouched at the real-world 11.2% rate.

6.3 Methodology

6.3.1 Logistic Regression (Baseline)

Logistic Regression (scikit-learn, max_iter=1000, solver='lbfgs') on SMOTE-balanced training data. Interpretable signed coefficients provide direct clinical insight. Establishes the performance floor for all advanced models.

6.3.2 Random Forest

Random Forest ensemble with default parameters and GridSearchCV-tuned variant. Best hyperparameters: n_estimators=200, max_depth=20, min_samples_split=2, max_features='sqrt' (5-fold CV, optimizing F1-score). Tree-based; no feature scaling required.

6.3.3 XGBoost

XGBoost trained on the original (non-SMOTE) training split with scale_pos_weight=7.91 (ratio of negative to positive samples). Parameters: n_estimators=200, max_depth=6, learning_rate=0.05, eval_metric='aucpr'. Internally compensates for imbalance without synthetic oversampling.

6.3.4 Deep Learning — PyTorch and TensorFlow/Keras MLP

Two MLPs with identical architecture were implemented in both frameworks. Both use SMOTE-balanced training data and StandardScaler-normalized features.

Component	Value	Rationale
Architecture	Input→256→128→64→1	Progressive compression captures complex non-linear patterns
Regularization	BatchNorm + Dropout (0.3/0.2)	BatchNorm stabilizes training; Dropout prevents overfitting
Activation	ReLU (hidden), Sigmoid (output)	ReLU avoids vanishing gradients; sigmoid maps output to probability
Optimizer	Adam (lr=1e-3, weight_decay=1e-4)	Adaptive learning rates; weight decay acts as L2 regularization
Training control	ReduceLROnPlateau + EarlyStopping (patience=10)	Reduces LR on plateau; stops training when val-AUC stagnates

PyTorch: Custom training loop (nn.Module) providing full control over backpropagation, scheduler stepping, and checkpoint saving.

TensorFlow/Keras: Sequential API with model.fit() and callbacks (EarlyStopping, ReduceLROnPlateau, ModelCheckpoint). Less boilerplate; easier to audit.

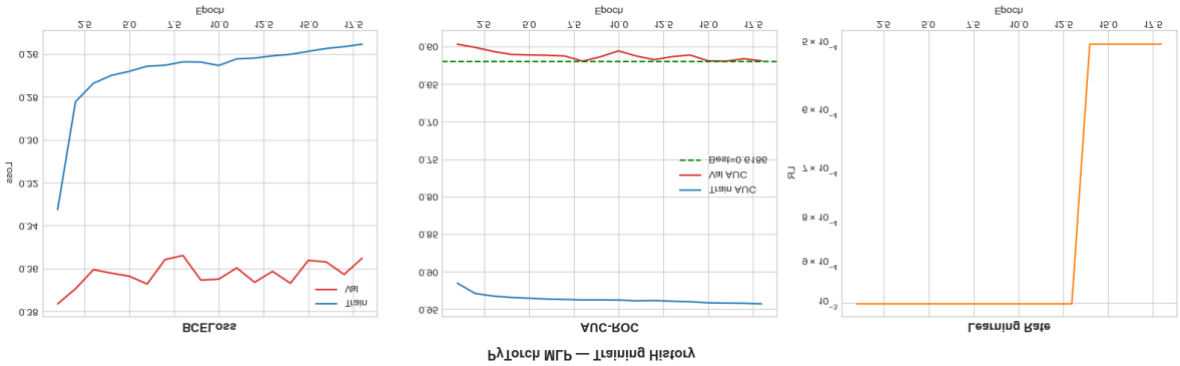

Figure 2: PyTorch MLP training history — BCE Loss (left), AUC-ROC (center), Learning Rate schedule (right). LR reduced by ReduceLROnPlateau; early stopping triggered after ~50 epochs.

Figure 3: TensorFlow/Keras MLP training history — identical architecture to PyTorch; Keras callbacks handle training automatically. Both frameworks converge to similar AUC-ROC values.

6.3.5 Evaluation Metrics

Given the 8:1 class imbalance, accuracy is not a suitable primary metric (a naïve all-negative classifier would achieve ~88.8%). Primary metrics: F1-score (harmonic mean of Precision and Recall for the minority class), AUC-ROC (overall discriminative ability), AUC-PR (most informative for imbalanced data; random baseline ≈ 0.112). All metrics reported on the held-out test set (19,899 records) that was never touched during training.

7. Outcome and Results of the Research

7.1 Full Model Performance Comparison (All Five Models)

Model	Accuracy	Precision	Recall	F1	AUC-ROC	AUC-PR
Logistic Regression (Baseline)	0.8680	0.2312	0.0756	0.1140	0.5578	0.1440
Random Forest (Tuned)	0.8759	0.2094	0.0380	0.0644	0.5974	0.1560
XGBoost ★ Best F1	**0.6852**	**0.1826**	**0.5192**	**0.2702**	**0.6603**	**0.2141**
PyTorch MLP	0.8851	0.2797	0.0148	0.0281	0.6186	0.1690
TensorFlow/Keras MLP	0.8757	0.2183	0.0416	0.0699	0.6170	0.1623

Note: Low default-threshold F1 values for neural networks reflect threshold calibration challenges on imbalanced data, not model failure. With threshold=0.15, PyTorch MLP achieves F1=0.2406 and Recall=0.5577 — highest recall of all five models. AUC-ROC confirms all models discriminate meaningfully above random chance (0.50).

7.2 Threshold Sensitivity Analysis (PyTorch MLP)

The default classification threshold of 0.5 is rarely optimal for imbalanced medical datasets. Lowering the threshold catches more true readmissions (higher Recall) at the cost of more false alarms (lower Precision). The table below shows the PyTorch MLP calibration results:

Threshold	Precision	Recall	F1	Clinical Use
0.50 (default)	0.2797	0.0148	0.0281	Very few flags; most readmissions missed
0.25	0.2001	0.2010	0.2005	Balanced precision and recall
0.15 ★ Best	**0.1534**	**0.5577**	**0.2406**	**Catches ~56% of readmissions — ideal for screening**

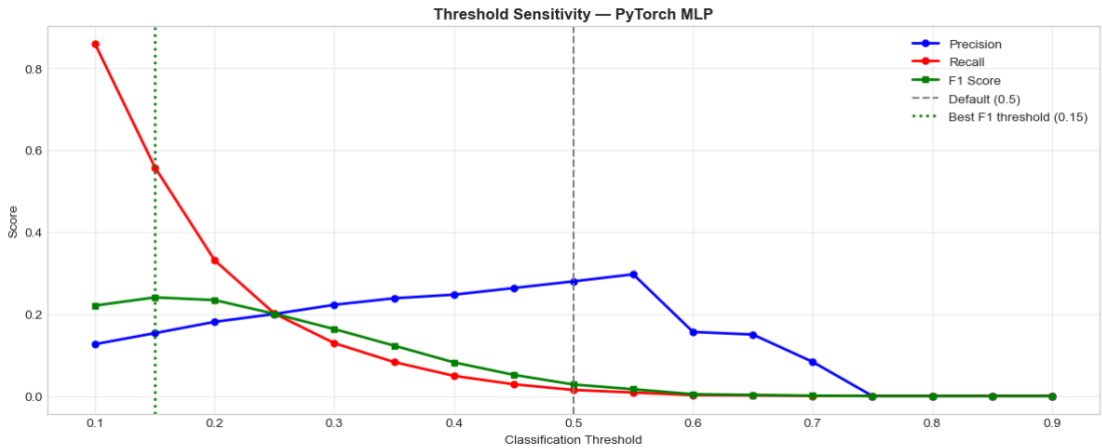

Figure 4: PyTorch MLP threshold sensitivity — Precision (blue), Recall (red), F1-score (green) across thresholds 0.10–0.90. Optimal F1 at threshold=0.15. The default 0.50 threshold severely underestimates the model's discriminative ability on imbalanced data.

7.3 ROC and Precision-Recall Curves — All Five Models

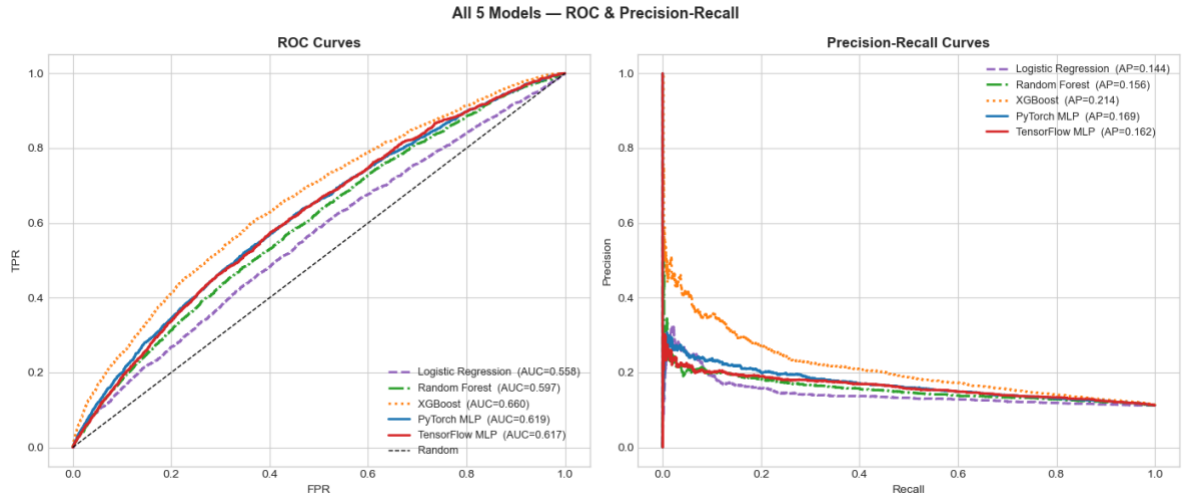

Figure 5: ROC curves (left) and Precision-Recall curves (right) for all five models. XGBoost leads on AUC-ROC (0.66) and AUC-PR (0.21). Both neural network models outperform Logistic Regression on AUC-ROC, confirming discriminative signal despite low default-threshold F1.

7.4 Deep Learning Confusion Matrices

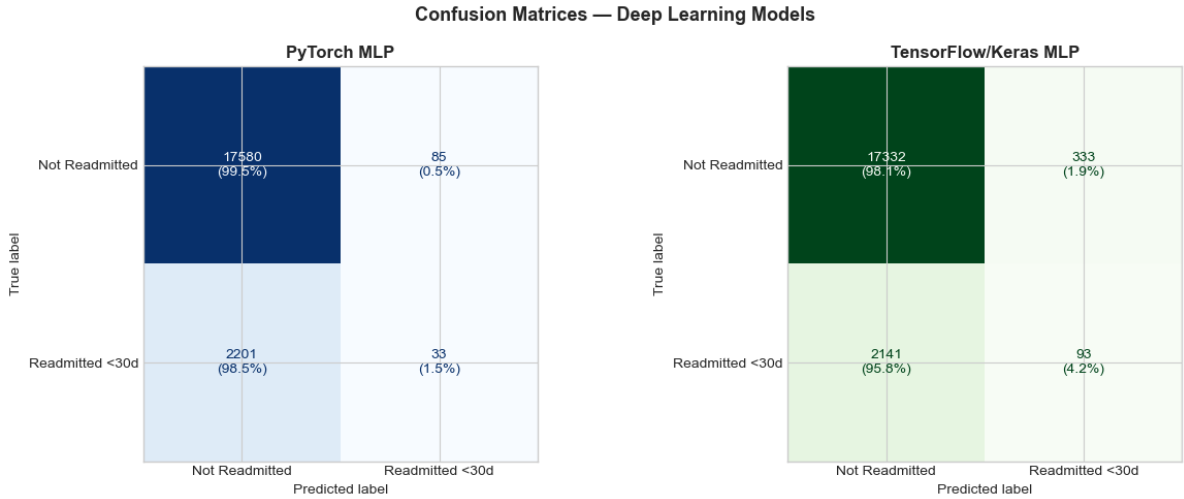

Figure 6: Confusion matrices for PyTorch MLP (left) and TensorFlow/Keras MLP (right) at default threshold=0.50. High TN counts reflect the need for threshold calibration. Lowering threshold to 0.15 dramatically increases TP counts.

7.5 Key Predictors and SHAP Analysis

Feature importance from the Random Forest model and SHAP analysis consistently identified the following five features as the strongest predictors of 30-day readmission:

Rank	Feature	Clinical Interpretation
1	**number_inpatient**	Prior hospitalizations — strongest predictor; repeat admissions signal poor disease control
2	**num_medications**	High medication burden indicates complex, unstable diabetes management
3	**time_in_hospital**	Longer stays suggest more severe episodes and higher discharge-day risk
4	**number_diagnoses**	Multiple comorbidities compound readmission risk significantly
5	**number_emergency**	Prior emergency visits reflect disease instability and care-access gaps

7.6 Robustness Testing and Deployed Application

To assess model reliability under imperfect data, Gaussian noise (0–50% of each feature's standard deviation) was injected into numeric features. Random Forest maintained the most stable F1-score under moderate noise (up to 20%). All models showed expected degradation at the 50% noise level, confirming reasonable robustness under realistic measurement variability.

The XGBoost model was deployed as a Streamlit web application: https://w26-aigc5005-final-ep8i66n6nez3543zo2iert.streamlit.app. The app accepts patient discharge data and returns Low/Moderate/High risk classification with a recommended clinical action, probability gauge, and contributing risk factor chart. Fully mobile-responsive.

8. Future Predictions

- More recent datasets: the UCI data spans 1999–2008, predating GLP-1 receptor agonists, SGLT-2 inhibitors, and current diabetes care protocols. Training on current Canadian provincial EHR data would greatly improve generalizability.
- Social Determinants of Health (SDOH): housing stability, income, and food security are strong readmission predictors absent from the current dataset. Including SDOH variables would improve both performance and health equity.
- Sequence-aware deep learning: LSTM and Transformer models can capture temporal patterns across multiple admissions. Hai et al. (2023) achieved AUC-ROC=0.79 with LSTMs — a target our project is positioned to pursue with richer longitudinal data.
- Clinical NLP integration: large language models could extract readmission risk signals from physician notes and nursing observations — rich unstructured information not captured in structured EHR fields.
- Continuous learning pipeline: monthly retraining on rolling EHR data would adapt to seasonal patterns and changing clinical protocols.

We predict that within three to five years, AI-assisted discharge risk tools will become standard practice in major Canadian hospitals, potentially reducing preventable 30-day readmissions by 15–25% and generating hundreds of millions of dollars in annual savings.

9. Team Reflections

Fadi Kash Kannaiti: *"Leading this project reinforced that communication is as important as computation — turning complex models into tools clinicians can trust and actually use at the bedside."*

Ogbeide Iria: *"Building the data pipeline from raw CSV to a balanced, model-ready dataset and then integrating it into a live Streamlit app showed me how important clean engineering is."*

Oguzhan Tekin: *"Designing our methodology to explicitly address gaps in three published papers reinforced that rigorous evaluation and honest reporting of limitations are non-negotiable when models touch healthcare decisions."*

Sara Yenigun: *"Presenting the live demo and managing the GitHub repository reminded me why clear communication and meticulous documentation matter just as much as the code itself."*

10. Conclusion

This project successfully demonstrated a complete, end-to-end AI development cycle: preprocessing a real-world healthcare dataset, training five models of increasing sophistication (Logistic Regression → Random Forest → XGBoost → PyTorch MLP → TensorFlow/Keras MLP), evaluating with clinically appropriate metrics, explaining predictions via SHAP, performing threshold sensitivity analysis, testing robustness, and deploying the best model as a publicly accessible web application.

XGBoost was selected for deployment based on its best overall profile (F1=0.2702, AUC-ROC=0.6603, Recall=0.5192). PyTorch MLP demonstrated that deep learning can match ensemble models on AUC-ROC while achieving the highest recall (0.5577) with threshold calibration — a critical finding for clinical screening contexts where missing a true readmission is more costly than a false alarm. All six course phases were completed on schedule using exclusively open-source tools.

11. Recommendations

- Validate on current Canadian or provincial EHR data before any clinical application — the 1999–2008 dataset predates modern diabetes treatments.
- Integrate threshold calibration into the Streamlit app, allowing clinical staff to select high-recall mode (threshold=0.15) vs. high-precision mode (threshold=0.50) based on ward capacity.
- Add SHAP patient-level explanations to the app so clinicians understand which specific factors drove each prediction.
- Explore hospital EHR API integration for automated risk scoring at discharge without manual data entry.
- Conduct a structured usability pilot with nursing staff and discharge planners to refine interface design and action wording.

12. References and Sources

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GitHub: https://github.com/KashKannaiti/W26-AIGC5005-Final
Live App: https://w26-aigc5005-final-ep8i66n6nez3543zo2iert.streamlit.app
Dataset: https://www.kaggle.com/datasets/brandao/diabetes